

Left over or opting out?

Squeeze, mismatch and surplus in Chinese marriage markets

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Abstract

Marriage is declining in China. Among singles, the probability of marrying in 2019 was half that of marrying in 1999. We estimate a [Choo and Siow](#) model using census data to quantify the relative roles of changes in population structure and changes in marital surplus, i.e., the value of marriage. We find that the increase in the supply of educated people explains half of the decline, partly due to a mismatch between highly-educated women and less-educated men. The deterioration of female-to-male ratio, known as marriage squeeze, explains an additional 13% for men. The decrease in surplus accounts for the remainder.

Keywords: Marriage markets, Sex ratio, Education, China

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1 Introduction

Marriage in China has been steadily declining for a decade. Policymakers are concerned by high rates of celibacy among young women and young men for several reasons. For women, celibacy contributes to low birth rates in a country where out-of-wedlock births are rare. Fertility rates have remained below replacement levels for more than thirty years, leading to serious demographic challenges (Dollar et al., 2020). For men, celibacy is associated with unhappiness, adverse health outcomes including premature deaths, and increased likelihood of criminal activity in many contexts (Zhou and Hesketh, 2017; Va et al., 2011; Edlund et al., 2013; Chang et al., 2021). The fact that people are choosing to marry later, or even not at all, therefore has broad implications (Shu and Chen, 2023).

A large, multidisciplinary literature aimed at understanding the reasons behind this decline has identified three key factors. The first factor is the *marriage squeeze*, which refers to the relative scarcity of women compared to men on the marriage market. The One-Child Policy combined with a strong preference for sons generated a substantial sex imbalance (Li et al., 2011). In the 2020 census, there were approximately 35 millions more men than women, fueling concerns about “leftover” men (*shengnan*) (National Bureau of Statistics of China, 2021). The second factor is the *marriage mismatch*, which occurs when singles struggle to find a suitable partner due to a shortage of their preferred type. Chinese traditions encourage female hypergamy; marriages in which the groom has less education than the bride are frowned upon. As education has risen faster for women than for men in recent decades, there is a shortage of suitable husbands for highly educated women, who end up as “leftover” women (*shengnu*). The third factor is the decrease in *marriage surplus*, namely the perceived benefits of being married versus being single. Changing social norms and preferences, greater gender equality in employment opportunities, as well as the increasing costs of having children, lead more and more individuals to “opt out” of marriage, at least temporarily.

In this paper, we quantify the role of each factor. We use the Choo and Siow (2006b) model, which allows us to identify and estimate the marital surplus using data on the number of singles and the number of matches by education category in a given year. We observe how many males and females of each category are single at the beginning of year t , whether they match during the year, and the category of their partner. This information can be constructed for the year 1999 using the 1% sample of the 2000 Chinese Census, and for the year 2019

combining aggregate statistics from the 2020 Chinese Census and survey data from the China Family Panel Studies. We proceed in three steps: (1) estimate the marital surplus in 1999 and 2019; (2) predict counterfactual marriages in 2019 under the assumption that the surplus remained constant at 1999 levels during the 20-year period; and (3) simulate different scenarios changing the population structure (the sex ratio and the level of education by gender) of new cohorts joining the market between 1999 and 2019. In line with the literature on decomposition methods (e.g. [Eika et al. \(2019\)](#)), our methodology is descriptive and abstracts from partial and general equilibrium considerations.

We find that the marital surplus indeed decreased between 1999 and 2019. We also find a reduction in assortative matching. The strongest variation in surplus (-56%) is observed for matches between less-educated men and less-educated women, the category that used to generate the highest surplus and was the most prevalent in 1999. Interestingly, matches between less-educated men and more-educated women, which used to generate the lowest surplus and were the least prevalent in 1999, are the only type of matches experiencing no reduction in surplus. This suggests that the preference for female hypergamy has weakened over time. When we simulate the evolution of the market holding surplus constant, we find that there would be 132 million singles in 2019 compared to 150 million in the real data.

In the counterfactual analysis, we are mainly interested in the annual matching rate, that is the probability of finding a match during the year for a single man and a single woman. This probability was roughly divided by two between 1999 and 2019 for both men and women. When we hold marital surplus constant at 1999 levels and update the flow of singles to reflect the observed composition of new cohorts, we find that 59% of the real decline in matching rates can be attributed to changes in population structure for men, with the remaining 41% being attributed to changes in surplus. For women, 48% is attributed to structure and 52% to surplus. Next, we vary the structure of incoming cohorts to investigate which dimensions matter for men and women. The marriage squeeze has a negative impact on male matching rates, contributing to the decline by 13%. The impact on female matching rates is positive, implying that in the absence of changes in education and marital surplus, single women would have been more likely to find a match in 2019 than in 1999. In addition, the rise in the number of educated individuals explains 45% of the male decline and 55% of the female decline. Changes in the supply of educated men and women may reduce matching rates for two reasons: first, educated people tend to marry later, and second, educated women do not tend to marry down. We find that

roughly two thirds of the contribution of education can be explained by the overall increase in education for both men and women, and one third by a mismatch between well-educated women and less-educated men. Finally, we implement the same decomposition separately in urban marriage markets and in rural marriage markets. We find that the contribution of population structure, in particular the sex ratio and the increase in overall education levels, is higher in rural markets, whereas the contribution of surplus is higher in urban markets, in particular for women.

Our article contributes to the literature on marriage markets in China by studying the main drivers of the decline in marriage rates in a unified, quantitative framework. In the literature, each factor tends to be studied separately by researchers in different disciplines using different methods. On the one hand, quantitative models are used to study the marriage squeeze. The evolution of the marriage market, taking into account the cohort sizes on both sides, is simulated. The goal is to estimate whether and when the imbalanced sex ratio has become a first-order factor. Projections depend on assumptions on age at marriage and spousal age gap. There is a broad consensus that the marriage squeeze matters and will peak in the 2040s-50s. Estimates for the most recent period (pre-2020) are mixed, with some papers arguing that sex ratios already make a difference ([Guilmoto, 2012](#)) whereas others argue that they do not ([Jiang et al., 2016](#)). The methodology is often very sophisticated, but the underlying model fails to account for changes in the value of marriage. On the other hand, qualitative methods are used to document the changing strategies employed by individuals in the market ([Qian and Qian, 2014](#); [You et al., 2016](#)). The key research question is to determine to what extent younger generations, and in particular women, are willing and able to reject marriage. Some studies emphasize that besides individual preferences and social norms, the pool of potential partners is an important factor influencing matches ([Li et al., 2017](#); [To, 2015](#); [Eklund and Attané, 2017](#); [Song et al., 2017](#)). However, as pointed out by [Schwartz \(2013\)](#), the methodology does not allow the relative importance of different factors to be quantified.

We also relate to the literature on matching models in economics. The methodology proposed by [Choo and Siow \(2006b\)](#) to decompose changes in marriage patterns into changes in population composition and changes in surplus has been applied in Canada ([Choo and Siow, 2006a](#)), in the US ([Cornelson and Siow, 2016](#); [Chiappori et al., 2017, 2020b](#)) and in the UK ([Chiappori et al., 2020a](#)). In the Chinese context, [Choo and Siow \(2006b\)](#) has been used to study the impact of the 1958-61 famine on the marital attractiveness of smaller cohorts ([Brandt et al., 2016](#)),

the marriage penalty for educated women (Brandt et al., 2018),¹ and the value of transmitting family name (Yang and Spencer, 2022). Chen and Choo (2023) use a dynamic version of the model to quantify the impact of the One-Child Policy on marriage distribution through its effect on the population and sex ratios. Other papers have studied the impact of the marriage squeeze on savings, marriage and intra-household allocations (Wei and Zhang, 2011; Porter, 2016; Ong et al., 2020). Our paper attempts to quantify how much of the decline in Chinese marriage can be attributed to changes in sex ratio, changes in education and changes in marital surplus. In particular, we isolate the role of the mismatch between highly-educated women and less-educated men, contributing to the nascent literature in economics studying hypergamy (Almås et al., 2023; Koh et al., 2025). In terms of methodology, we propose a simple way to simulate the evolution of counterfactual markets over time by explicitly modeling the evolution of the stock of singles. By dealing with the sequential entry of new cohorts, we overcome one limitation of the purely static version of the Choo and Siow (2006b) model. This allows us to answer counterfactual questions such as: what would have happened if the marital surplus had remained constant over the past 20 years?

Beyond the Chinese context, the decline in marriage has been observed throughout East and South East Asia (Jones, 2017). Like China, some of these countries experienced deep changes in attitudes toward marriage as well as a deterioration of sex ratios at birth (e.g. Vietnam) and a rapid expansion of education (e.g. Japan, South Korea, Singapore, and more recently Thailand and Vietnam). Our quantification exercise can easily be replicated to analyze the evolution of marriage outside China.

2 Methodology

2.1 Intuition

Choo and Siow (2006b) propose an empirical model with transferable utility and no search friction: everyone can meet any potential partner, access all pertinent information, and transfer utility to their partners without any restriction. The main advantage is to identify the gains from matching using only data on matches and without requiring data on transfers between

¹The analysis of Brandt et al. (2018) is the most closely related to our paper although their focus is different. They investigate whether educated women face a marriage penalty in urban China and conclude that they do not. Their main goal is to estimate gender-specific gains to marriage (the part of the surplus captured by each spouse) and to describe the evolution between 2000 and 2008/09 for various education groups. In contrast, we are mainly interested in *counterfactual* analysis to understand what would have happened if trends in population structure (not only education but also sex ratio) had been different.

partners, which are typically unobserved.

Denote μ_{ij} the number of matches between a man of type i and a woman of type j observed in the data. μ_{i0} and μ_{0j} are the number of men of type i and women of type j , respectively, who remain unmatched. Denote π_{ij} the *marriage surplus*, defined as the total systematic gain to marriage per partner for a pair (i, j) relative to the total systematic gain per partner from remaining unmarried. The key result is the following relationship:

$$\pi_{ij} = \ln\left(\frac{\mu_{ij}}{\sqrt{\mu_{i0}\mu_{0j}}}\right) \quad (1)$$

The intuition behind this result is the following: if we observe many marriages between type i and type j , these unions must be valuable. However, it can also be that there are many men of type i and many women of type j . We need to control for the scale dimension in order to isolate the value dimension. We use the number of unmatched individuals of both types to do so. The value of a match is high if these matches are often observed *relative to* the frequency of types in the population. The identification relies on assumptions regarding the distribution and the separability of unobserved heterogeneity in matching payoffs.

2.2 Set-up in Choo and Siow (2006b)

In this section, we provide more details on the modeling of the matching market, and the identification and interpretation of π_{ij} . This is a summary of Choo and Siow (2006b) without new insights.

Let us suppose there are I types of men and J types of women, where each type is defined by the intersection of observable characteristics.² The number of types, denoted by I and J , is finite. Each individual belongs to a specific type (and only one). A matching function $\mu(M, F : \Pi)$ relates the marriage distribution μ to population vectors M and F , and to marriage surplus Π . μ is represented as a matrix $(I + 1) \times (J + 1)$, where each element μ_{ij} indicates the number of matches or the number of unmatched individuals. This matching function is deemed “feasible” if all individuals in the market are matched with either 0 or 1 partner, and “stable” if no matched individual would prefer to be single, and if no pair of individuals would prefer to be matched together instead of with their current partner.

Feasibility requires the matching function to satisfy the following conditions:

²In the original framework, the types are constrained to be discrete; then, some extensions have allowed for continuous types.

$$\mu_{i0} + \sum_{j=1}^J \mu_{ij} = m_i, \forall i, \quad (2)$$

$$\mu_{0j} + \sum_{i=1}^I \mu_{ij} = f_j, \forall j, \quad (3)$$

$$\mu_{0j}, \mu_{i0}, \mu_{ij} \geq 0, \forall i, j. \quad (4)$$

where m_i is the number of men of type i and f_j is the number of women of type j .

The utility of man s of type i marrying a woman of type j , V_{ijs} and the utility of a woman v of type j marrying a i -type man, V_{ijv} , are given by:

$$V_{ijs} = \alpha_{ij} - \tau_{ij} + \epsilon_{ijs} \quad (5)$$

$$V_{ijv} = \gamma_{ij} + \tau_{ij} + \epsilon_{ijv}$$

The utility consists of three parts: (1) a gross systematic payoff α_{ij} or γ_{ij} , which is exogenous and depends only on the types of both spouses; (2) a (possibly negative) transfer τ_{ij} from i -type men to j -type women, which is determined endogenously at equilibrium; (3) an individual-specific random component ϵ_{ijs} or ϵ_{ijv} , which captures unobserved characteristics/preferences of individual s or v .

If they stay single, individuals receive a type-specific payoff as well as an individual-specific payoff. The singlehood payoffs for the i -type man s and the j -type woman v are given by:

$$V_{i0s} = \alpha_{i0} + \epsilon_{i0s} \quad (6)$$

$$V_{0jv} = \gamma_{0j} + \epsilon_{0jv}$$

Thus, individuals will choose according to:

$$V_{is} = \max_y \{V_{i0s}, \dots, V_{ijs}, \dots, V_{iJs}\} \quad (7)$$

$$V_{jv} = \max_x \{V_{0jv}, \dots, V_{ijv}, \dots, V_{Ijv}\}$$

There are two important assumptions on the unobserved heterogeneity components ϵ . First, conditional on (i, j) , the joint surplus created by a match between a i -type man and a j -type woman does not depend on interactions between their unobserved characteristics. In other words, the match can happen because the unobservable characteristics of this specific woman make her attractive for all men of group i or because this specific man has strong unobserved preferences for all women of group j . What is ruled out by assumption in the

model is that the match happens because this specific man has unobserved preferences for unobserved characteristics of this specific woman. This is known as the *separability* assumption. Second, the ϵ terms are independent and identically distributed according to the Type I Extreme Value distribution (McFadden, 1973). This parametric assumption imposes restrictions on the patterns of substitution between the different choices of partners. Galichon and Salanié (2022) show that the second assumption can be relaxed.

Knowing the distribution of ϵ , we can deduce the quasi demand and supply for all types on the market. Choo and Siow (2006b), building on earlier work by McFadden (1973), show that the quasi demand equation by men of type i for women of type j is given by:

$$\ln \mu_{ij}^d = \ln \mu_{i0}^d + \alpha_{ij} - \alpha_{i0} - \tau_{ij} \quad (8)$$

where $\ln \mu_{ij}^d$ is the demand for j -type women by i -type men and $\ln \mu_{i0}^d$ is the number of unmarried i -type men. The quasi demand decreases with τ_{ij} . Similarly, they show that the quasi supply equation of women of type j for men of type i is given by:

$$\ln \mu_{ij}^s = \ln \mu_{0j}^s + \gamma_{ij} - \gamma_{0j} + \tau_{ij} \quad (9)$$

where $\ln \mu_{ij}^s$ is the supply of j -type women for i -type men and $\ln \mu_{0j}^s$ is the number of unmarried j -type women. The quasi supply is increasing with τ_{ij} .

The marriage market clears when the demand for women by men is equal to the supply of women for men for all types, given the equilibrium transfers τ_{ij} . That is, for all i, j , $\mu_{ij} = \mu_{ij}^d = \mu_{ij}^s$. Using this condition and summing the last two demand and supply equations, we get:

$$\ln \mu_{ij} - \frac{\ln \mu_{i0} + \ln \mu_{0j}}{2} = \frac{\alpha_{ij} - \alpha_{i0} + \gamma_{ij} - \gamma_{0j}}{2} \quad (10)$$

If we denote $\pi_{ij} = \frac{\alpha_{ij} - \alpha_{i0} + \gamma_{ij} - \gamma_{0j}}{2}$, we obtain the key result presented in equation 1. At equilibrium, this simple equation links the marriage surplus to the number of newlywed couples divided by the geometric average of men and women who stay unmarried. Choo and Siow (2006b) conclude that this ratio of observable marriage market outcomes is a sufficient statistic for quantifying the quality of marriage matches.

π_{ij} has a clear interpretation: it captures the per-capita systematic net gains to marriage (relative to remaining single) for a couple consisting of an i -type man and j -type woman. $\pi_{i,j}$

is determined by exogenous preference parameters (α, γ) and is unaffected by changes in the quantities of men and women of different types. What does depend on the population vector is the transfer between spouses, which guarantees that the demand for a spouse of a given type equals the supply. Allowing for *spillover effects* is an important contribution of this model compared to previous quantitative models of the marriage market: matching outcomes between i -type men and j -type women are affected not only by the quantities of i -type men and j -type women but also by the quantities of men and women of all other types.

2.3 From observed matches to estimation of surplus

In our application, we suppose that we have two types, H and L . We need to observe (i) the population vector, i.e. the number of singles of each type at the beginning of the period: m_H , m_L , f_H and f_L ; and (ii) the number of matches in all four categories: μ_{HH} , μ_{LH} , μ_{HL} and μ_{LL} . We can then infer the number of unmatched individuals by type. In practice, this means filling in the population matrix; then, we use the formula to estimate the surplus matrix, as illustrated below.

$$\begin{array}{c}
 \text{Women} \\
 \begin{array}{cc}
 \left(\begin{array}{cc} \mu_{HH} & \mu_{HL} \\ \mu_{LH} & \mu_{LL} \end{array} \right) & \begin{array}{c} \mu_{H0} \\ \mu_{L0} \end{array}
 \end{array}
 \xrightarrow{\pi_{i,j} = \ln\left(\frac{\mu_{i,j}}{\sqrt{\mu_{i0}\mu_{0j}}}\right)}
 \begin{array}{cc}
 \left(\begin{array}{cc} \pi_{HH} & \pi_{HL} \\ \pi_{LH} & \pi_{LL} \end{array} \right) & \\
 \mu_{0H} & \mu_{0L}
 \end{array}
 \end{array}$$

2.4 From estimated surplus to simulation of matches

Once we have estimated the surplus in a given market, we can perform simulations by keeping the surplus constant (in blue) and changing the population vector (in red). This exercise requires solving the following system of eight equations and eight unknowns, where i and j take values H and L .

$$\begin{array}{lcl}
 (1) & \mu'_{iH} + \mu'_{iL} + \mu'_{i0} = \textcolor{red}{m'_i} & \\
 (2) & \mu'_{Hj} + \mu'_{Lj} + \mu'_{0j} = \textcolor{red}{f'_j} & \\
 (3) & \mu'^2_{ij} = \mu'_{i0}\mu'_{0j}\exp(2\pi_{i,j}) &
 \end{array}
 \xrightarrow{\text{CS algorithm}}
 \begin{array}{cc}
 \left(\begin{array}{cc} \mu'_{HH} & \mu'_{HL} \\ \mu'_{LH} & \mu'_{LL} \end{array} \right) & \begin{array}{c} \mu'_{H0} \\ \mu'_{L0} \end{array} \\
 \mu'_{0H} & \mu'_{0L}
 \end{array}$$

Choo and Siow (2006a) propose a simple algorithm to solve this system (which also works when there are more than two types). The algorithm predicts the number of matches in all four categories and the number of unmatched individuals by type. We can then answer the question: what would happen if surplus stayed constant while population structure changed?

2.5 From simulated matches to construction of counterfactual markets

In order to study the evolution of the counterfactual marriage market over several years, we need to model the evolution of the stock of singles. This requires an extra step which is not discussed in Choo and Siow (2006a,b).

At period 0, we estimate the surplus, π_0 , using the observed number of matched and unmatched individuals. At the beginning of the next period, the number of singles, Singles_1 , is the sum of previously unmatched individuals and the new cohort entering the market. We simulate matches in period 1 using Singles_1 and π_0 , and we predict the number of unmatched individuals at the end of the period. We can iterate the process for every period using the following loop:

$$(\text{Unmatched}_{t-1} + \text{New Cohort}_t = \text{Singles}_t) \text{ combined with } \pi_0 \xrightarrow{\text{CS algorithm}} \text{Unmatched}_t$$

Singles_t has two components: (i) an endogenous component, the *stock* of individuals who were already on the market and did not match in the previous year; its structure (size, sex ratio, type, age) is an outcome determined by the surplus and population vectors in previous years; (ii) an exogenous component, the *flow* of individuals entering the market in that year as they become adults; the structure of this incoming cohort (size, sex ratio, type) changes over time for reasons unrelated to the marriage market.

With data on the marriage market in 1999 and 2019 as well as data on incoming cohorts during the 2000-2019 period, we can (i) take the observed matches in 1999 and estimate the surplus in 1999, (ii) simulate matches year-by-year holding surplus constant and updating the singles vector with simulated values for the stock and observed values for the flow, and (iii) compare the final simulated matches and the matches observed in 2019. This allows us to decompose the evolution into changes in the structure of incoming cohorts (captured by the counterfactual market) and changes in surplus (the residual explanation of the gap between the counterfactual market and the observed market).

To our knowledge, the final step in our methodology is new and allows us to answer counterfactual questions that cannot be addressed with the purely static version of Choo and

[Siow \(2006b\)](#). Without modeling the dynamic process, we can only answer questions such as: what would happen tomorrow if we suddenly change the surplus? Or what would happen tomorrow if we remove the current population of singles and replace it with a different one? These questions deal with a prospective counterfactual: they take a given market, change one component and predict the new matches. In contrast, we are interested in a retrospective counterfactual. We want to know what would have happened in 2019 if the surplus had stayed constant during the previous 20 years, not what would happen in 2020 if the surplus suddenly switched back to 1999 levels. We also want to know what would have happened in 2019 if the cohorts entering the market during the 2000-2019 period had the same sex ratio and/or education levels as the cohort who entered the market in 1999, not what would happen in 2020 if the pool of singles was suddenly replaced by the pool of singles in 1999. These retrospective questions are harder to answer because they require modelling the sequential entry of new cohorts and the accumulation of singles over time.

2.6 Assumptions and limitations

There are two main limitations generated by key assumptions in [Choo and Siow \(2006b\)](#). First, there is no search friction. Any change in meeting opportunities between groups or in dating technologies, which influence match frequencies in real life, is captured by the matrix π . In the structural model, π identifies the net gains to marriage. However, when interpreting the evolution of π over time, it is important to keep in mind that they can potentially be driven by changes in search frictions (see the discussion in [Eika et al. \(2019\)](#)). In section 5.1, we explain why this is unlikely to be the case in our context. Second, there is no anticipation. When deciding to match in period t , individuals do not take into account the possibility of delaying marriage until the next period nor the possibility of divorce and remarriage later. [Choo \(2015\)](#) extends the model to include dynamic considerations such as strategic delays and potential remarriage opportunities. In practice, compared to the static version, the dynamic model requires larger datasets, is more computationally intensive, and is sensitive to assumptions about the way individuals form expectations about future marriage markets. We show that the dynamic estimates of the surplus are similar to the static estimates in Section 5.1.

3 Data

3.1 Data sources

We use three data sources. First, the 1% sample of the 2000 National Population Census ([Minnesota Population Center, 2020](#)) provides information on *who married whom* in 1999. We combine microdata on current marital status and year of first marriage to identify who was single at the beginning of 1999, who married during the year and with whom. We observe the number of male and female singles by education category, and the number of matches, by education of *both* spouses. We are therefore able to construct (i) the vector of singles, (ii) the matching rate, and (iii) the frequency of match types in 1999. We observe around 55,000 marriages in 1999, which allows us to have very reliable estimates of the characteristics of matches.

Second, microdata from the 2020 National Population Census are not publicly available yet. Instead, we use aggregate statistics available in the China Population Census Yearbook 2020 ([National Bureau of Statistics of China, 2020](#)) to infer *who married* in 2019, but we cannot know with whom.³ By observing the number of male and female singles, by education category, and the number of matches, we can construct (i) the vector of singles and (ii) the matching rate in 2019.

Third, in order to construct the frequency of match types in 2019, we use the 2020 China Family Panel Studies (CFPS). This is a household survey that provides, along with other data, the relevant information (marital status and year of marriage). We observe around 150 marriages in 2019, which is enough to get estimates of the frequency of each match type.⁴ Note that unlike the 2020 census which covers 100% of the population, CFPS is representative of 95% of the Chinese population: specific provinces like Tibet and Inner Mongolia are not included. Also, the CFPS sample size is 1/1000 of the census 1% sample. This is why we use CFPS data only to estimate the surplus in 2019 and to discuss the evolution of surplus over time. We prefer to use the 2020 census to compute the characteristics of the incoming cohorts and to implement the counterfactual analysis.

³The aggregates are official statistics based on a 10% sample of the total population, who completed the long questionnaire covering detailed questions about marital status and marriage age.

⁴We checked that the vectors of singles and the matching rates are nearly the same when we estimate them using census data and using CFPS data. We also checked that the frequency of match types remains the same if we pool marriages observed in 2018 and 2019 to increase the CFPS sample size.

3.2 Definitions

We define the incoming cohort in year t as the cohort of new adults, i.e. those turning 18 during the previous year. Although the minimum legal marriage age is 20 for women and 22 for men, we observe some marriages at age 19 in the data. Using the legal age to define the pool of potential partners would therefore raise issues when we count the number of matches.

Apart from age, we make one restriction when defining the pool of singles.⁵ We exclude individuals who have been previously married (divorced or widowed) because these marriage markets are quite segmented in China and the policy concern is about first marriages.⁶

Next, we define the type as *Low* (L) if the individual has less than high school education, and *High* (H) if the individual has ever attended high school. High school typically starts at age 16 so the decision to enroll or not is taken before entering the marriage market. Types are therefore predetermined; this is why we choose high school rather than college. The incoming cohorts during the 2000-2019 period are born between 1981 and 2000. We use the 2020 aggregate statistics to obtain information on the number of individuals by gender and education in each cohort.

Finally, we focus on 2019 because marriage rates after 2020 were exceptionally low due to covid-19 and our goal is to capture long-term trends. In principle, we could have used the 2010 National Population Census to add a point in 2009 but the 0.2% sample we could access was not representative.

3.3 Marriage market in 1999

We construct the marriage market in 1999 using the 1% sample of the 2000 Census. We restrict the sample to individuals aged at least 19 in 2000, hence were adults (18 and older) at the beginning of 1999. The marriage market in 1999 consists of individuals who are single at the start of 1999; in 2000, these individuals are either still single or newlywed couples who married in 1999 and 2000. We identify the latter group by exploiting information on the year of the first marriage. We classify individuals as matched in 1999 if they married during that year, and as unmatched if they married in 2000 or remained single in 2000.

To estimate the marital surplus in 1999, we need to know the number of matches by education

⁵In addition, we restrict the sample to native Chinese given that marriage markets are segmented and foreigners account for a tiny fraction of the Chinese population (less than 0.5% of the marriage market in 2019).

⁶Among marriages happening in 2019, 92% are first marriages for both spouses. Among single men (women) in 2020, 12% (9%) were divorced and 11% (30%) were widowed. Over 99.5% of singles aged 70 and older remain single.

of both spouses. We combine (i) information on demographic characteristics for everyone living in the same household and (ii) information on each member’s relationship to the household head in order to identify the couples within a household. We proceed in three steps. First, we match the household head with the spouse of the household head and identify 34,257 couples who married in 1999. Second, we match the child of the household head with the child-in-law of the household head. We identify 20,851 additional couples who married in 1999. Third, we match the parents of the household head; however, and not surprisingly, none of these couples married in 1999. With this procedure, we have 55,108 matched couples, which account for 70% of the 78,513 men and 77,803 women who married in 1999 in the sample. We checked that the educational distributions in our sub-sample of matched couples closely resembles the distributions in the entire sample of men and women who married in 1999. We use the sub-sample of matched couples to compute the distribution of the four types of matches in 1999.

The remaining unmatched couples correspond to the following situations. First, a couple does not live in the same household; this is relatively rare, e.g. 9% of married household heads do not have a spouse in the household. Second, the couple resides in a collective household (larger dwelling units with several individuals who are not related by family links); this is also rare, e.g. about 3% of individuals aged 19 and above live in collective households and only one-third among them are married. Third, household heads have multiple children or children-in-law; this is the main reason why we cannot identify the couples correctly. We tried to increase the number of matched couples by making assumptions on who is married with whom among the multiple children.⁷ However, we ended up with a sub-sample of matched couples that was less comparable to the entire sample of couples in terms of education. We therefore believe that maximizing the fraction of couples matched can be counterproductive in terms of representativeness. Our main assumption in constructing the distribution by type of marriage is that couples who can be properly matched with our 3-step procedure are representative of all couples.

We obtain the following numbers for the marriage market in 1999, expressed in millions of individuals (see the step-by-step procedure in [Appendix A](#) for more details).

⁷This is the approach of [Brandt et al. \(2018\)](#). For instance, they match individuals within a household based on marriage duration or they draw "replacement" spouses from the age-education distribution observed among the matched couples.

	Women	
Men	$\begin{pmatrix} 1.5 & 0.7 \\ 0.4 & 5.2 \end{pmatrix}$	18.4
		46.0
	12.8	21.5

3.4 Marriage market in 2019

We construct the marriage market in 2019 using the aggregated statistics of the 2020 Census. The marriage market in 2019 consists of those who married in 2019 and 2020, as well as those reported as single in 2020. The census aggregates report the number of singles in 2020 and the number of individuals married in 2019 and 2020 by gender. Hence, we can calculate the number of matched and unmatched men and women.

The aggregated data of the 2020 Census does not allow us to identify the characteristics of the spouse. Instead, we turn to the 2020 wave of the China Family Panel Studies (CFPS) for this purpose. Once again, we restrict the sample to those aged 19 and above in 2020, who reported being single in 2020 or entered their first marriage in 2019 and 2020. Those married in 2019 are considered as being matched on the 2019 market. Unlike census data, we do not have to identify the couples in CFPS: the survey directly lists the spouse's attributes for all individuals. We can then compute the distribution of the types of matches among those couples who married in 2019. We use the CFPS distribution to infer the total number of couples of each type formed in 2019.

We obtain the following numbers for the marriage market in 2019, expressed in millions of individuals (see the step-by-step procedure in [Appendix A](#) for more details).

	Women	
Men	$\begin{pmatrix} 3.1 & 0.7 \\ 0.8 & 1.1 \end{pmatrix}$	57.2
		36.0
	45.9	10.4

4 Descriptive Statistics

4.1 Outcome of interest: Matching rates

Based on the marriage market matrices, we calculate the annual matching rate as the fraction of individuals who married during the year among those who were single at the start of the year. In 1999, we obtain $\frac{7.8}{7.8+18.4+46.0} = 0.108$ for men, and $\frac{7.8}{7.8+12.8+21.5} = 0.185$ for women. In 2019, we obtain $\frac{5.7}{5.7+57.2+36.0} = 0.057$ for men, and $\frac{5.7}{5.7+45.9+10.4} = 0.091$ for women. Therefore, the annual matching rate among singles declined by 47% for men and by 51% for women between 1999 and 2019.

The annual matching rate is closely related to the distribution of ages at marriage. For instance, if the rate is equal to λ and constant between age 19 and 30, the proportion who marries before age 30 is equal to $1 - (1 - \lambda)^{11}$. The decrease in matching rates observed in the data would therefore correspond to a decrease in the proportion of individuals married before age 30 from 72% to 48% for men, and from 89% to 65% for women. The difference in levels between men and women reflects the fact that men typically marry at an older age than women, and therefore are in the marriage market for a longer time. We are not seeking to explain the gender difference in levels, but rather the trends in both male and female matching rates.

In the next sections, we discuss three often-cited explanations for the decrease and provide some descriptive statistics supporting them.

4.2 Explanation 1: Imbalanced sex ratios

The right panel of Figure 1a plots the evolution of the sex ratio by cohort using the 2020 census aggregates. For cohorts active in the marriage market in 1999 (born in 1980 and before), the sex ratio is stable, around 1.03 men per woman. For the incoming cohorts, born between 1981 and 2000, we can distinguish two periods: before 1990, the sex ratio remains relatively balanced, whereas after 1990, the sex ratio becomes more and more imbalanced over time, reaching an all-time high of 1.13 men per woman. We therefore expect the male matching rate to decrease between 1999 and 2019, and in particular after 2010, as imbalanced cohorts enter the market. In the counterfactual exercise, we modify the sex ratio of incoming cohorts to quantify the contribution of the marriage squeeze.

4.3 Explanation 2: Education expansion

The left panel of Figure 1a plots the evolution of the share of individuals who ever attended high school by gender and cohort. Two features of the evolution are striking: first, education increased dramatically over the past decades, and second, the gender gap reversed. In the 1970 cohort, 21% of women and 26% of men have ever attended high school. In the 2000 cohort, these proportions are 81% and 75%, respectively. Women have overtaken men in terms of education in the cohorts born in the late 1980s and onwards. This implies that the composition of cohorts active on the marriage market is very different in 1999 (the average education is low, and lower for women) and in 2019 (the average education is high, and higher for women).

This change in the supply of educated men and women may affect the matching rates in two ways. First, high- and low-educated people may have different values of marriage *in general*: they marry later and have different outside options. Second, they have different preferences in terms of partner type. In particular, in the presence of female hypergamy, the female advantage in education generates a mismatch between highly-educated women and less-educated men. A first hint that female hypergamy exists is that when looking at the distribution of marriage types in 1999, we find that marriages between L -type men and H -type women are very rare: they account for only 5% of all marriages. The opposite combination, between H -type men and L -type women, is twice as common (9.5%). However, these proportions reflect both the population structure and the marital surplus. We use our methodology to test whether LH matches indeed generate a lower surplus than HL matches. Then we consider counterfactual scenarios in which the education of incoming cohorts grew less, or grew equally for men and women.

4.4 Explanation 3: Decrease in the value of marriage

Before turning to the estimation results, we illustrate changes in the value of marriage relative to being single using data from the China General Social Survey. In particular, one question asks whether the respondent agrees with the statement that “Even the worst marriage is better than singlehood.” The survey was conducted in 2006 and respondents were between 20 and 60 years old. Figure 1b plots the share of men and women agreeing with the above statement, by cohort. For older cohorts, the levels are high, around 40%, indicating that the outside option of remaining single used to have a very low value for a substantial share of people. The trend is

clearly downward: for younger cohorts, less than 30% agree with the statement.⁸

If we do the same analysis by education level, we find that highly-educated individuals agree less than relatively low-educated individuals for both men and women, and the decline over time is particularly pronounced for highly-educated women. Therefore, in the incoming cohorts, we expect the value of marriage to decline over time, because of composition effects – the percentage with high school education increases – and because of changes in the value of marriage within each group. By estimating the model, we get a better measure of the marital surplus in the sense that (i) the measure has a structural interpretation and (ii) the measure is match-specific.

5 Results

5.1 Estimation

In Figure 2, we report the surpluses in 1999 and 2019 calculated using equation 1 (see matrices in Appendix A for more details). Note that all surpluses are negative, meaning that the average value from matching is lower than the average value from remaining single for both partners. This is because in a given year, most people remain single. Only the individuals who derive a large idiosyncratic payoff from a given match do choose to match; the majority do not.⁹

In 1999, the ranking of the different types of matches is: $LH < HL < HH < LL$. This indicates that (i) there is a strong preference for assortative matching; (ii) among assorted couples, low educated types derive more value from matching than high educated types; (iii) there is a strong distaste for female hypogamy.

In 2019, the surpluses generated by the most prevalent matches, LL and HH , have decreased by 56% and 22%, respectively. The strongest decline is observed among low-educated individuals, which challenges the common wisdom that opting out of marriage is a trend specific to highly-educated groups. Moreover, the surplus generated by LH has not decreased: it remains stable over time, almost closing the gap with the surplus of HL , which decreased by 9%. This evolution suggests that the strong distaste for women marrying down attenuates as the female advantage in education grows. Our finding is consistent with recent qualitative research arguing that norms around hypergamy are not experienced as strongly by younger cohorts (Eklund, 2018). One

⁸Note that we cannot separate cohort and age in this dataset; we emphasize the cohort interpretation to be consistent with qualitative studies documenting generational gaps rather than life cycle trends. Unfortunately, we do not have information for the youngest incoming cohorts (born between 1986 and 2000).

⁹In most applications, including the seminal analysis by Choo and Siow (2006a) and the analysis of Chinese marriage markets by Brandt et al. (2018), estimated surpluses are negative.

explanation is that parents, who used to put pressure on their daughters to marry up, are now mainly worried that their daughters may not marry at all.

If we compute the assortativeness index introduced by [Chiappori et al. \(2020b\)](#), we find that assortative matching decreased over time. The SEV (Separable Extreme Value) index, computed as the difference between the sum of on-diagonal surpluses and the sum of off-diagonal surpluses, is positive and smaller in 2019 than in 1999. The advantage of this approach is to provide a clear theoretical interpretation in terms of supermodularity of the surplus function (complementarity between spouses' characteristics).

As explained above, the disadvantage is to rely on the assumption of frictionless matching. In principle, changes in search frictions could also affect matching patterns. However, in this context, frictions have probably decreased over time with the expansion of college education and the introduction of online dating technologies. [Ming and Luo \(2025\)](#) argue that China's college expansion reduced search frictions within the local marriage market created by colleges, thus facilitating marriage between highly educated spouses. In our framework, this would be captured by an increase in π_{HH} . Using data from Korea, [Lee \(2015\)](#) finds that online dating increases assortative sorting along education, while reducing sorting by geographical proximity. In our framework, this would imply that all π increase (higher meeting probability for all types) and that π_{HH} and π_{LL} should increase more than π_{HL} and π_{LH} (more assortative matching on education). These predictions are at odds with our findings: an overall reduction in surpluses and a reduction in assortativeness. For this reason, and in line with the sociology literature, we emphasize an interpretation in terms of changes in preferences rather than changes in search frictions.

Moreover, we investigate whether allowing for anticipation affects our conclusions. To estimate the dynamic model proposed by [Choo \(2015\)](#), we need to take a stand on how individuals form expectations. Under perfect foresight, they anticipate perfectly future changes in population structure and marital surpluses. This assumption is hard to (i) justify in contexts experiencing radical transformations such as contemporary China and to (ii) operationalize when the econometrician does not observe future markets. Alternatively, under myopic expectations, individuals believe that the market is in a steady state, meaning that population structure and surplus remain constant over time. Under this assumption, we estimate the 1999 surplus and report the results in [Appendix C](#).¹⁰ Figure [C.1a](#) plots the

¹⁰We cannot estimate the 2019 surplus because we observe too few marriages in the CFPS data.

distribution of marital surpluses for the four match types: HH , HL , LH , and LL . Unlike the static model, which has one value per match type, the dynamic model has several values per match type corresponding to different age groups, and that is why we use box plots. We find the same ranking as in the static model: $LH < HL < HH < LL$.¹¹

Overall, the analysis of surpluses shows that the value of marriage has changed in ways that is sometimes unexpected and impossible to infer from descriptive statistics. Changes in surplus should explain part of the decline in marriage. Next, we turn to the counterfactual exercise to quantify precisely how much.

5.2 Counterfactual number of singles

We start by simulating the evolution of the stock of singles: we fix the surplus at 1999 levels and, in each year t , we allow individuals born in $t - 19$ to join the market.¹² Figure 3 plots the evolution over 20 years and compares the final simulated value with the real value observed in 2019. If marital surplus had stayed constant during the 20-year period, there would be 18 million fewer singles in 2019, a drop by 12% compared to the real number. The structure would also be different: there would be 10 million fewer H-type men, 3 million *more* L-type men, 4 million fewer H-type women and 7 million fewer L-type women.

As expected, changes in surplus contributed to raising the total number of singles, as more men and women decided to *opt out*. However, a countervailing force was at play for L-type men and H-type women: the attenuation of the distaste for female hypogamy, which made LH matches relatively more likely to happen than in 1999.

5.3 Counterfactual matching rate

The analysis of the stock of singles indicates that even if marital surplus had stayed constant, the number of singles would have increased substantially, from roughly 100 million in 1999 to 130 million in 2019. This is because incoming cohorts are more numerous and have a different structure in terms of sex ratio and education. In order to focus on the role of the structure, we turn to a statistics that does not vary with the size of the population: the matching rate.

¹¹Figure C.1b plots the results disaggregated by spousal age difference. Again, the ranking across education types remains the same, irrespective of the age difference. Therefore, we do not seem to lose important information when we aggregate all age groups.

¹²We take into account the possibility that some of the previously unmatched individuals die. We calculate the death rates among singles by gender using the 2000 census: the annual death rate is equal to 0.3% for single men and 0.08% for single women.

We consider three simulations and, in each scenario, we predict the counterfactual decline in matching rate. First, we fix the surplus at 1999 levels, and we simulate the evolution of the market using the true sex ratio and the true education rates of the incoming cohorts. By comparing this first counterfactual matching rate and the true initial rate in 1999, we can estimate how large the decline would have been if the surplus had not changed, and only the population structure had changed. Second, we use the true education rates, but instead of the true sex ratio, we attribute to the incoming cohorts the sex ratio among cohorts active on the 1999 market (1.03). By comparing this second counterfactual matching rate and the true initial rate in 1999, we can estimate how large the decline would have been if neither the surplus nor the sex ratio had changed, and only education had changed. Third, we use the true education rates for men, but we change the education rates for women to reflect the gender gap among cohorts active on the 1999 market (4p.p. in favor of men) as well as the sex ratio. By comparing this third counterfactual matching rate and the true initial rate in 1999, we can estimate how large the decline would have been if neither the surplus nor the sex ratio nor the gender gap in education had changed, and only the overall level of education had changed.

The results are shown in Figure 4; we provide details in the figure note and we report the corresponding matrices in Appendix B. Figure 4a plots the counterfactual decline in matching rates in each simulation for men and women. The real decline in matching rates observed between 1999 and 2019 is shown at the extreme left. Figure 4b presents a waterfall plot to visualize how much each component contributes to explaining the real decline. We can obtain the contribution of changes in surplus by comparing simulation 1 and the observed decline; the contribution of changes in sex ratio (squeeze) by comparing simulations 1 and 2; the contribution of changes in educational gender gaps (mismatch) by comparing simulations 2 and 3; the contribution of changes in overall educational levels by looking at simulation 3.

In simulation 1, we find that the matching rate would have decreased by 28% for men and by 24% for women if surpluses had stayed constant. Given that the true decline is 47% for men and 51% for women, the change in marital surplus explains 41% and 52% of the change in matching rate for men and women, respectively. The remaining part is due to the change in population structure. Changes in population explain a larger share of the decline for men than for women because the squeeze contributes in opposite ways, as we can see in simulation 2. We find that if sex ratios had stayed constant, matching rates would have decreased by only 21% for men (compared to 28% in simulation 1) while they would have decreased by 28% for

women (compared to 24% in simulation 1). The deterioration of sex ratios therefore amplifies the decline for men, contributing to 13% of the observed change, whereas it attenuates the decline for women. If we compare simulation 2 and the observed decline, we learn about the role of education, which is the only component that we did not fix in the simulation. It turns out to be an important component for both men and women, explaining 45% ($=21/47$) and 55% ($=28/51$) of the observed decline, respectively. Finally, simulation 3 allows us to separate the contribution of the general increase in education from the mismatch between highly-educated women and low-educated men. If education had increased but the gender gap had stayed constant in favor of men, the matching rate would have dropped by 14% for men and by 19% for women. Comparing these numbers with simulation 2, we conclude that the mismatch explains 16% of the decline for both men and women. The remaining part, explaining 29% and 38% of the decline for men and women respectively, is due to the overall increase in education. In other words, the total contribution of education can be split into two thirds for the absolute increase for both men and women and one third for the relatively faster increase for women compared to men.

5.4 Extensions

Heterogeneity by urban-rural market. So far, we have considered the country as a whole as a unique marriage market. However, the literature has emphasized important differences between urban and rural areas (Shu and Chen, 2023; Meng, 2012). We repeat our analysis separately for cities and towns on the one hand and for villages on the other hand, using the classification of places of residence in the census. The results are valid under the assumption that both markets are separate and determined at age 19. Individuals may move before turning 19 (typically leaving rural areas to start college education in urban areas) but afterwards, we assume that they live and marry in the same market. Descriptively, if we look at the evolution of incoming cohorts, we observe that the sex ratio deteriorated much more in the rural market, while both markets experienced a strong increase in education and a reversal of the gender gap.

Results of the counterfactual analysis are reported in Figure 5. The decline in marriage has been stronger in the rural market (-56%) than in the urban market (-43%). In the urban market, the striking result is that changes in surplus explain 80% of the decline for women: fixing the surplus at 1999 levels is enough to reduce the decline from 43% to 9%. If we fix the educational gender gap on top of the surplus, there is no decline anymore. Therefore, narratives emphasizing that educated women opt out of marriage seem to be particularly relevant in the

urban market. For urban men, the decomposition is more balanced, roughly similar to the decomposition in the whole sample, except that marriage squeeze barely matters. In the rural market, the predominant component is the change in population structure, and in particular the absolute increase in education; this alone explains 36% of the decline for men and 52% for women. Even if we fix surplus, sex ratio and educational gender gaps, we still predict a decline by 20% for men and 29% for women. Squeeze matters a great deal for rural men, contributing to 27% of the decline, as much as changes in surplus.

Defining type by *hukou* instead of education. The results of the decomposition exercise are specific to our definition of types as “ever attended high school”. Another partition of the population would potentially lead to a different conclusion regarding the contribution of changes in the population structure. In that sense, the contribution of the surplus is a residual and an upper bound: we explain 59% (resp. 48%) of the decline in marriage for men (resp. women) using only one dimension of the population vector. Another potentially interesting dimension would be *hukou* (agricultural or non-agricultural). Unfortunately, there are two data limitations. First, aggregate data from the 2020 census are not available by *hukou* status; therefore, we can only exploit the CFPS survey data which has fewer observations. Second, the *hukou* status is observed at the time of the survey, not at the time of marriage. If people change their status upon marriage, we mis-classify them. Keeping these caveats in mind, and using CFPS data, we find that the share of individuals with an agricultural *hukou* increased between the 1981 cohort and the 2000 cohort, and is persistently higher for men. This is consistent with higher fertility and a more imbalanced sex ratio in rural areas, where most agricultural *hukou*-holders live. Nonetheless, the magnitudes are small and the counterfactual exercise assigns a limited role to changes in *hukou* composition in explaining the change in marriage rates. We conclude that “high school attendance” is by far the most relevant dimension to define types in this context.¹³

6 Conclusion

Our results help explain the decline in annual marriage rates over the past two decades in China. The decline is not only driven by an excess of men in general, but also more specifically by an excess of less-educated men. Less-educated men are left over and highly-educated women opt out partly because they do not find a suitable partner. Looking ahead, the role of marriage

¹³We also considered defining type by ethnicity. However, changes along this dimension make little difference in the aggregate given that Han Chinese account for more than 90% of the population.

squeeze may become more prominent in the coming years as the accumulation of imbalanced cohorts over time increasingly impacts the pool of singles. On the other hand, mismatch may become less influential as preferences evolve, and the option of marrying someone with a lower level of education may become more acceptable for women.

In terms of policy implications, each factor can potentially be affected by interventions. First, any policy trying to rebalance sex ratios should reduce the scope of the marriage squeeze in the long run. The male-to-female ratio at birth has started to decline since the relaxation of the One-Child Policy in the mid-2010s, although it remains higher than the natural sex ratio at birth. The decline is particularly fast in rural areas and the urban-rural gap has almost disappeared in 2020 ([Tang and Hou, 2024](#)). The importance of “leftover” men should therefore decrease at some point, but this will be in the distant future as we need to wait until the most imbalanced cohorts have exited the market. Second, the recent reversal of the gender gap in education has been observed not only in China but in several middle to high-income countries, and policymakers are increasingly concerned about male school dropouts ([Reeves, 2022](#)). Empirical evidence on interventions targeting young men at risk is still rare; promising examples include financial incentives, educational curriculum that meets boys’ needs and interests (e.g. through a mixture of technical skills and internships), lifelong learning programs to help adult men to continue their education, youth service programs, and mentoring programs with role models ([Welmond and Gregory, 2022](#)). Policies addressing male educational underachievement could play a key role in solving the issue of mismatch on the marriage market. Third, policymakers can raise marital surplus using fiscal tools and social benefits. For instance, joint taxation may be designed in a way that favors couples over singles, and the allocation of scarce public resources (e.g. public housing) may give priority to couples over singles. However, marital surplus captures much more than the economic tradeoff between living alone and cohabiting. Is the worst marriage better than singlehood? The answer depends on whether individuals are willing and able to remain single, and policymakers have only a limited influence on these factors.

This application illustrates the potential of the methodology developed by [Choo and Siow \(2006b\)](#) to deepen our understanding of marriage markets. The methodology is computationally simple to implement and does not require detailed data. We suggest two avenues for future research: (i) applying it more systematically to quantify the drivers of marriage rates in different countries; (ii) explaining the evolution of surplus over time by considering, for instance, the role of changes in fertility preferences, divorce rights, housing markets or labor markets.

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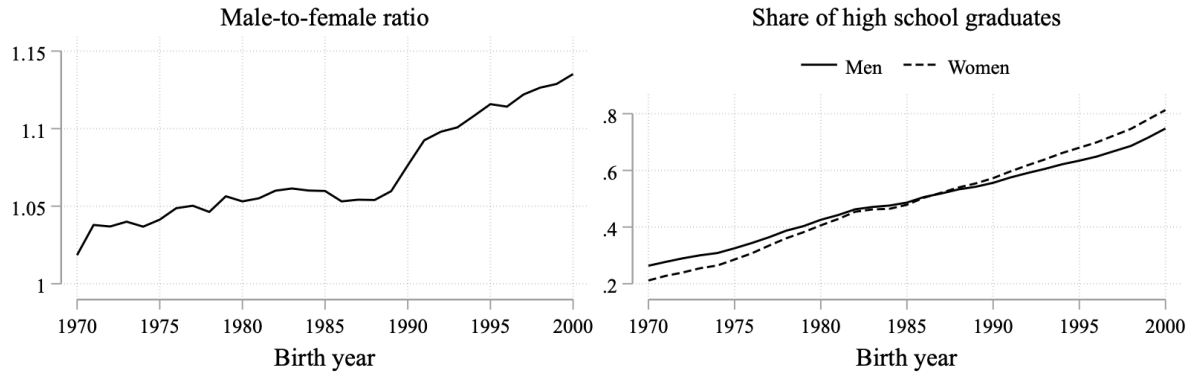
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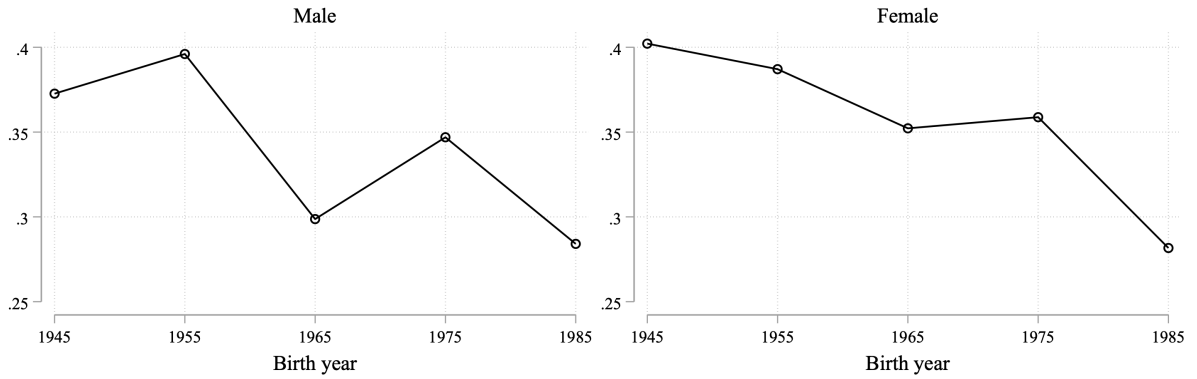
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Figure 1: Population structure and attitudes toward marriage, by cohort

(a) Population structure

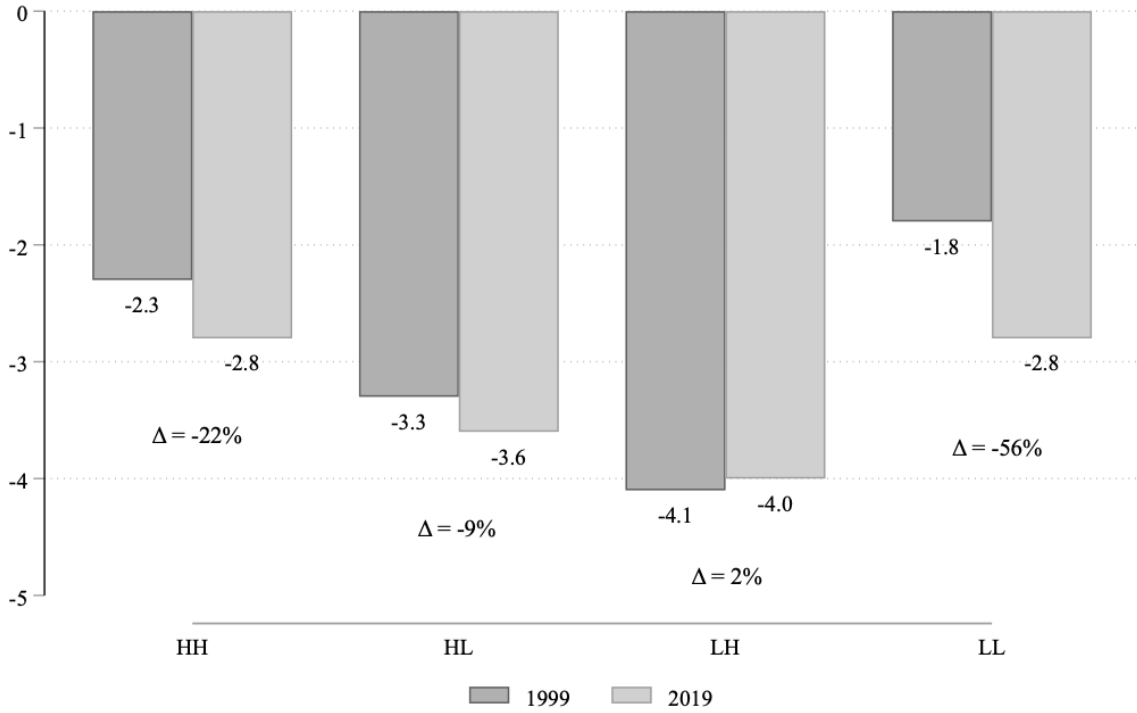


(b) % agree that "Even the worst marriage is better than singlehood" in 2006



Note: Figure (a) shows the male-to-female ratio (left panel) and share of high school educated by gender (right panel) by birth year, calculated with the aggregated statistics from the 2020 Census. Figure (b) shows the share of male respondents (right panel) and female respondents (left panel) agreeing with the statement "Even the worst marriage is better than singlehood" in CGSS 2006. The dot for birth year t represents the average over all individuals born between $t - 9$ and t . For example, the number for birth year 1945 represents the average over individuals born between 1936 and 1945. Cohorts active on the marriage market in 1999 are born in 1980 and before. Cohorts entering the market between 1999 and 2019 are born between 1981 and 2000.

Figure 2: Estimated marriage surplus, by marriage type

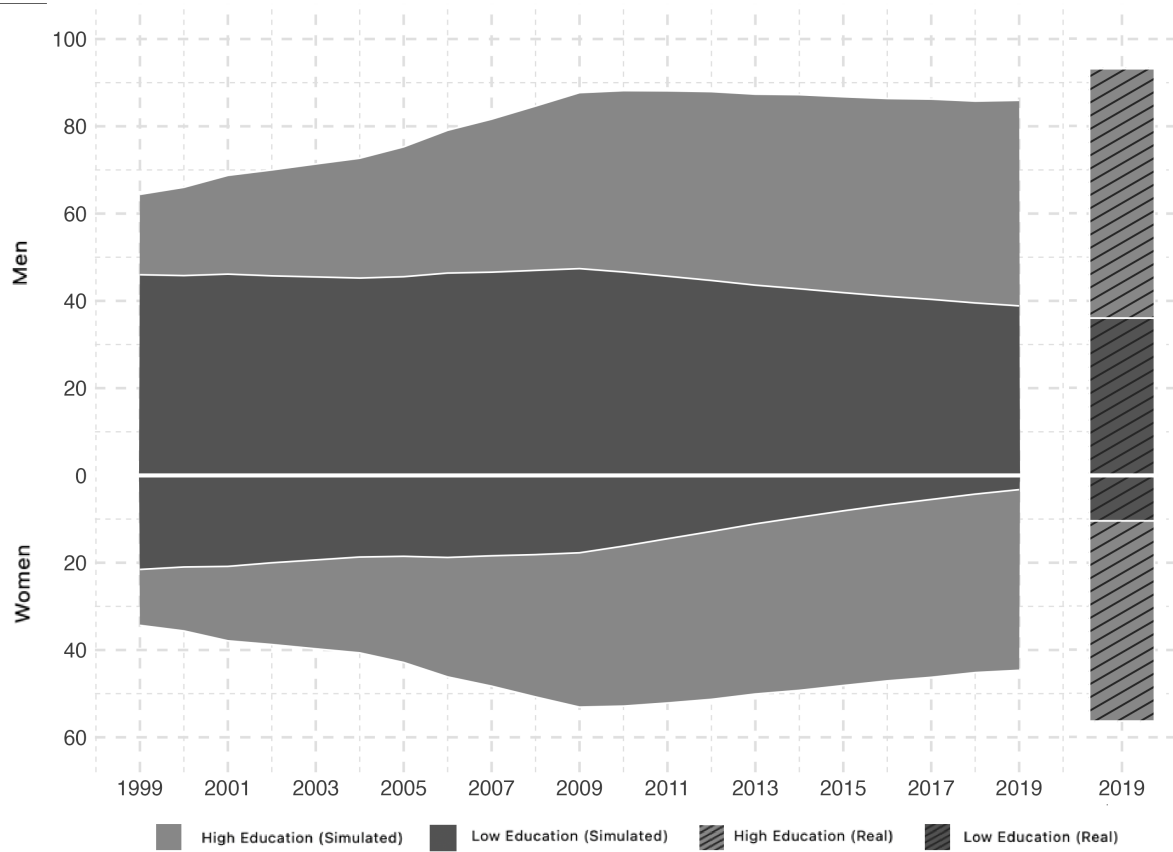


Note: The figure shows the *marriage surplus* π_{ij} in 1999 and 2019, defined as the total systematic gain to marriage per partner for a pair (i, j) relative to the total systematic gain per partner from remaining unmarried.

- *HH*: marriages between *H*-type men and *H*-type women (19% of marriages in 1999 and 55% in 2019)
- *HL*: marriages between *H*-type men and *L*-type women (10% of marriages in 1999 and 12% in 2019)
- *LH*: marriages between *L*-type men and *H*-type women (5% of marriages in 1999 and 13% in 2019)
- *LL*: marriages between *L*-type men and *L*-type women (66% of marriages in 1999 and 20% in 2019)

H-type means “ever attending high school” and *L*-type means “never attending high school”. See Section 2 and Appendix A for details on the methodology.

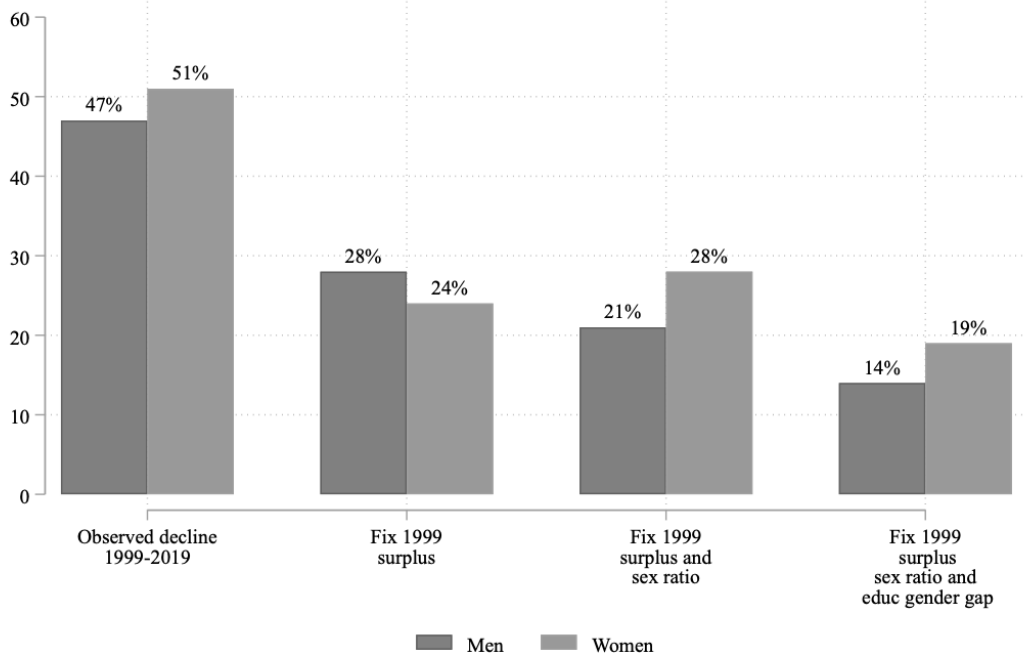
Figure 3: Number of singles in the counterfactual market keeping surplus constant at 1999 levels



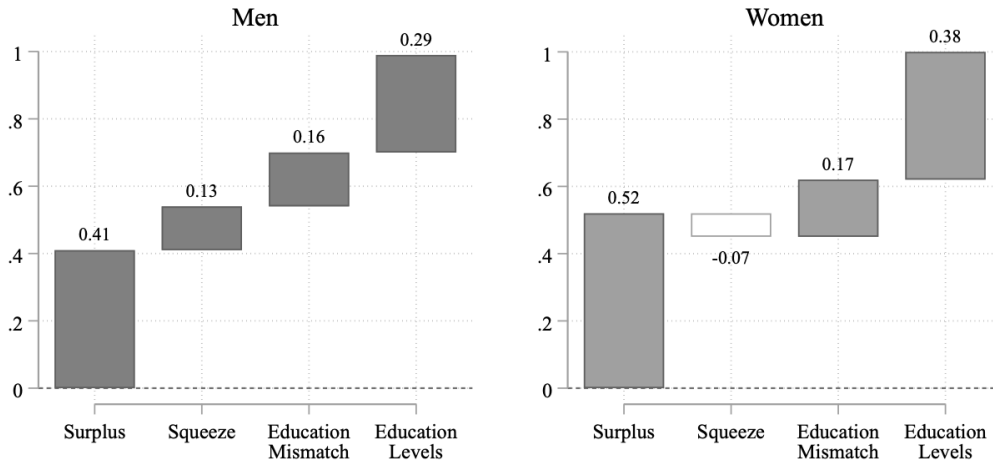
Note: The figure plots the evolution of the number of singles in millions, by gender and education type, in the counterfactual market. We keep the surplus constant at 1999 levels and simulate the evolution of the market year-by-year. The bar at the extreme right represents the real number of singles in 2019. Note that the distribution of singles by educational levels is only available for the year 2020; we construct a proxy of the real vector of singles in 2019 using the distribution of education in the observed vector of singles in 2020. See Section 2 and [Appendix B](#) for details on the methodology.

Figure 4: Counterfactual decline in matching rates between 1999 and 2019

(a) Counterfactual decline in matching rates fixing different factors



(b) Fraction of the real decline between 1999 and 2019 explained by each factor



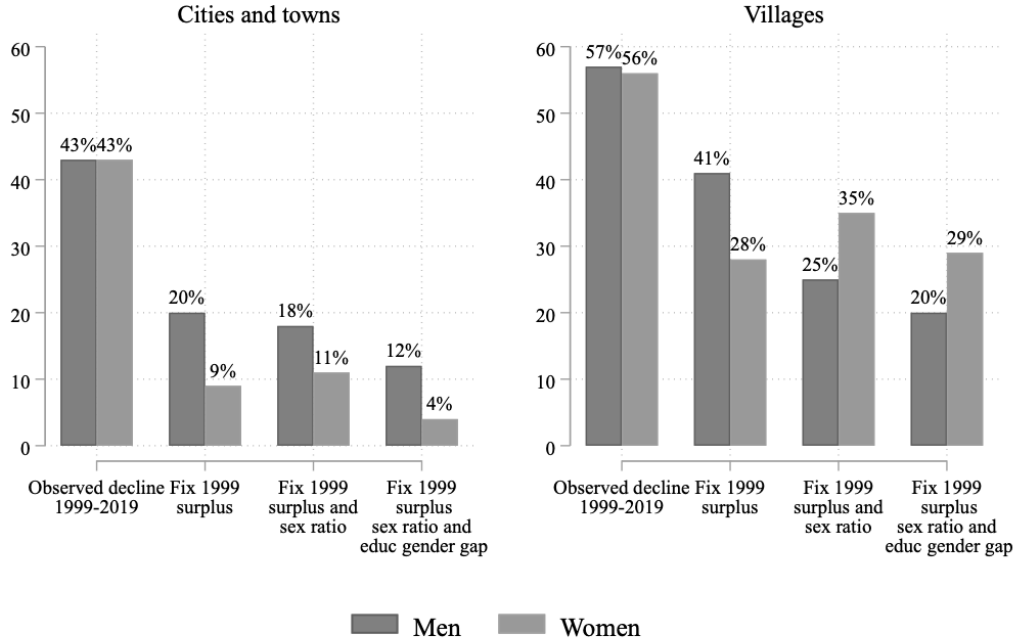
Note: See Section 2 and Appendix B for details on the methodology. Each counterfactual market is characterized by the following vector of incoming cohorts and matching surplus:

	% <i>H</i> -type men	% <i>H</i> -type women	Sex ratio	Surplus
Base market 1999	0.299*	0.255*	1.03*	π_{1999}
Real market 2019	Men_{t-19}	$Women_{t-19}$	$Cohort_{t-19}$	π_{2019}
1: Fix surplus	Men_{t-19}	$Women_{t-19}$	$Cohort_{t-19}$	π_{1999}
2: Fix surplus and sex ratio	Men_{t-19}	$Women_{t-19}$	1.03*	π_{1999}
3: Fix surplus, sex ratio, and education gender gap	Men_{t-19}	$Men_{t-19}-0.044^*$	1.03*	π_{1999}

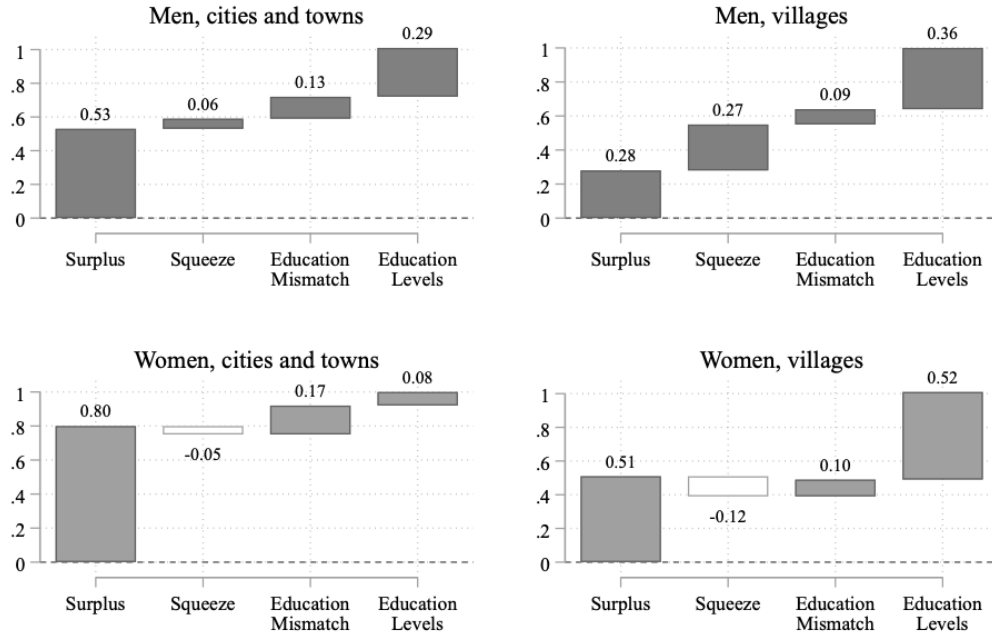
* mean of cohorts active in the 1999 market (born in 1966-1980)

Figure 5: Counterfactual decline in matching rates in urban and rural markets

(a) Counterfactual decline in matching rates fixing different factors



(b) Fraction of the real decline between 1999 and 2019 explained by each factor



Note: See notes below Figure 4 for more details. We use the same methodology separately in cities and towns and in villages. Base markets had the following characteristics:

	% H -type men	% H -type women	Sex ratio	Surplus
Base market 1999 - cities and towns	0.448 ^u	0.406 ^u	1.02 ^u	π_{1999}^u
Base market 1999 - villages	0.106 ^r	0.064 ^r	1.06 ^r	π_{1999}^r

^u mean of cohorts active in the 1999 market (born in 1966-1980) in cities and towns

^r mean of cohorts active in the 1999 market (born in 1966-1980) in villages

Appendix

Appendix A Real markets

Marriage market in 1999

Step 1: Number of singles by types and number of matches observed in the 1% sample of the 2000 census

	Women	
Men	$\left(78,513 \text{ men and } 77,803 \text{ women} \right)$	184,216
		459,834
	128,149	215,405

In the 1% sample of the 2000 census, we observe 644 thousand men who were single at the beginning of the year (they are either single at the time of the census or they got married in 2000); 184 of them have high school education and 460 have not. Similarly, we observe 343 thousand women who were single at the beginning of the year, 128 with high school education, 215 without. All these individuals were active on the market in 1999 and did not find a partner in 1999. In addition, we observe 78,513 men and 77,803 women who got married in 2019. Among them, we can only match 55,108 couples. For the others, we cannot identify their spouses in the household (see detailed discussion in the main text). We use the sample of matched couples to infer the distribution of match types.

Step 2: Number of matches by types in 1999 observed in the sample of matched couples in the 1% sample of the 2000 census, expressed in units and in fraction of the total number of matched couples in parenthesis

	Women	
Men	$\left(\begin{array}{cc} 10,644 (= 19\%) & 5,214 (= 10\%) \\ 2,822 (= 5\%) & 36,428 (= 66\%) \end{array} \right)$	

In the sample of matched couples obtained in step 1 (N=55,108), 19% are between high education men and women, 66% between low education men and women, 10% between high education men and low education women, and 5% between low education men and high education women.

Step 3: Estimated number of matches, expressed in millions of units

$$\begin{array}{cc} & \text{Women} \\ \text{Men} & \begin{pmatrix} 1.5 (= 7.8 \times 0.19) & 0.7 (= 7.8 \times 0.10) \\ 0.4 (= 7.8 \times 0.05) & 5.2 (= 7.8 \times 0.66) \end{pmatrix} \end{array}$$

According to step 1, we know that approximately 7.8 millions of couples were formed in 1999 (we multiply the number of observations in the 1% sample by 100 to get a estimate of the total population). We use the proportions obtained in step 2 to infer the total number of couples, by type.

Step 4: Estimated market, expressed in millions of units, and calculation of the 1999 surplus

$$\begin{array}{cc} & \text{Women} \\ \text{Men} & \begin{pmatrix} 1.5 & 0.7 \\ 0.4 & 5.2 \end{pmatrix} \end{array} \begin{array}{cc} 18.4 & \\ 46.0 & \end{array} \xrightarrow{\pi_{i,j} = \ln(\frac{\mu_{i,j}}{\sqrt{\mu_{i0}\mu_{0j}}})} \begin{pmatrix} -2.3 & -3.3 \\ -4.1 & -1.8 \end{pmatrix}$$

12.8 21.5

We combine the number of couples obtained in step 3 and the number of singles obtained in step 1 (multiplying by 100) to construct the full matrix.

Marriage market in 2019

Step 1: Number of singles by types observed in the aggregate statistics from the 2020 census

$$\begin{array}{cc} & \text{Women} \\ \text{Men} & \begin{pmatrix} 573,762 \text{ men and } 556,564 \text{ women} \end{pmatrix} \end{array} \begin{array}{cc} 5,715,147 \\ 3,604,682 \end{array}$$

4,588,267 1,041,032

The aggregates are official statistics based on a 10% sample of the total population. In the sample, 9.3 million men were single at the beginning of 2020 (either single at the time of the census or they got married in 2020); 5.7 of them have high school education and 3.6 have not.¹⁴

¹⁴We only know the education distribution of individuals by marital status but not by year of marriage. Therefore, we use the education distribution of single men and women at the time of the census, between October and December 2020, to infer the education distribution of single men and women in the beginning of 2020.

Similarly, 5.6 million women were single at the beginning of the year, 4.6 with high school education, 1.0 without. All these individuals were active on the market in 2019 and did not find a partner in 2019. In addition, 574 thousand men and 557 thousand women got married in 2019.

Step 2: Number of matches by types in 2019 observed in the CFPS data, expressed in units and in fraction of the total number of matches in parenthesis

$$\begin{array}{cc} & \text{Women} \\ \text{Men} & \begin{pmatrix} 82 (= 55\%) & 18 (= 12\%) \\ 20 (= 13\%) & 30 (= 20\%) \end{pmatrix} \end{array}$$

In the CFPS sample of couples (N=150), 55% are between high education men and women, 20% between low education men and women, 12% between high education men and low education women, and 13% between low education men and high education women.

Step 3: Estimated number of matches, expressed in millions of units

$$\begin{array}{cc} & \text{Women} \\ \text{Men} & \begin{pmatrix} 3.1 (= 5.65 \times 0.55) & 0.7 (= 5.65 \times 0.12) \\ 0.8 (= 5.65 \times 0.13) & 1.1 (= 5.65 \times 0.20) \end{pmatrix} \end{array}$$

According to step 1, we know that approximately 5.65 millions of couples were formed in 2019 (we multiply the number of observations in the 10% sample by 10 to get a estimate of the total population). We use the proportions obtained in step 2 to infer the total number of couples, by type.

Step 4: Estimated market, expressed in millions of units, and calculation of the 2019 surplus

$$\begin{array}{cc} & \text{Women} \\ \text{Men} & \begin{pmatrix} 3.1 & 0.7 \\ 0.8 & 1.1 \end{pmatrix} \end{array} \begin{array}{cc} 57.2 & \\ 36.0 & \end{array} \xrightarrow{\pi_{i,j} = \ln(\frac{\mu_{i,j}}{\sqrt{\mu_{i0}\mu_{0j}}})} \begin{pmatrix} -2.8 & -3.6 \\ -4.0 & -2.8 \end{pmatrix}$$

45.9 10.4

We combine the number of couples obtained in step 3 and the number of singles obtained in step 1 (multiplying by 10) to construct the full matrix.

Appendix B Simulated markets

We start with the number of unmatched individuals at the end of 1999 and we fix π at the 1999 level (see numbers in [Appendix A](#)). The incoming cohort in year 2000 is born in 1981. This cohort has 9.8 million men and 9.3 million women, a sex ratio (SR) of 1.05 and a share of high school graduates equal to 44.2% for men and 42.8% for women (see [Figure 1a](#)). Among cohorts active in the 1999 market (born in 1966-1980), the average sex ratio is 1.03 and the average share of high school graduates is 29.9% for men and 25.5% for women (hence a gender gap of 0.044 in favor of men). We consider the following three simulations for the incoming cohort in year 2000:

	Nb men	Nb women	% H men	% H women	SR	Surplus
1: Fix surplus	9.8	9.3	0.442	0.428	1.05	π_{1999}
2: Fix surplus and SR	9.8	9.5	0.442	0.428	1.03	π_{1999}
3: Fix surplus, SR and educ gender gap	9.8	9.5	0.442	0.398	1.03	π_{1999}

After running the CS algorithm, we obtain the number of unmatched individuals at the end of 2000. We repeat the same exercise incorporating new cohorts one by one.

Simulation 1: fix surplus

Numbers for the incoming cohort are observed in the 2020 census aggregates and expressed in millions of individuals.

$$\begin{aligned}
 (1) \quad & \mu'_{HH} + \mu'_{HL} + \mu'_{H0} = 18.4 + 4.3 (= 9.8 \times 0.442) \\
 (2) \quad & \mu'_{LH} + \mu'_{LL} + \mu'_{L0} = 46.0 + 5.5 (= 9.8 \times (1 - 0.442)) \\
 (3) \quad & \mu'_{HH} + \mu'_{LH} + \mu'_{0H} = 12.8 + 4.0 (= 9.3 \times 0.428) \\
 (4) \quad & \mu'_{HL} + \mu'_{LL} + \mu'_{0L} = 21.5 + 5.3 (= 9.3 \times (1 - 0.428)) \xrightarrow{\text{CS algorithm}} \begin{pmatrix} 1.7 & 0.8 \\ 0.4 & 5.1 \end{pmatrix} \begin{matrix} 20.2 \\ 45.8 \end{matrix} \\
 (5) \quad & \mu'_{HH}{}^2 = \mu'_{H0}\mu'_{0H} \exp(2 \times -2.3) \qquad \qquad \qquad 14.7 \quad 21.0 \\
 (6) \quad & \mu'_{HL}{}^2 = \mu'_{H0}\mu'_{0L} \exp(2 \times -3.3) \\
 (7) \quad & \mu'_{LH}{}^2 = \mu'_{L0}\mu'_{0H} \exp(2 \times -4.1) \\
 (8) \quad & \mu'_{LL}{}^2 = \mu'_{L0}\mu'_{0L} \exp(2 \times -1.8)
 \end{aligned}$$

Simulation 2: fix surplus and sex ratio

We keep the total number of men constant as observed in the census and we change the number of women to reflect the sex ratio (1.03) among cohorts active in the 1999 market (born in 1966-80). We keep the fraction of H -type as observed in the census.

$$\begin{aligned}
(1) \quad & \mu'_{HH} + \mu'_{HL} + \mu'_{H0} = 18.4 + 4.3(= 9.8 \times 0.442) \\
(2) \quad & \mu'_{LH} + \mu'_{LL} + \mu'_{L0} = 46.0 + 5.5(= 9.8 \times (1 - 0.442)) \\
(3) \quad & \mu'_{HH} + \mu'_{LH} + \mu'_{0H} = 12.8 + 4.1(= 9.5 \times 0.428) \\
(4) \quad & \mu'_{HL} + \mu'_{LL} + \mu'_{0L} = 21.5 + 5.4(= 9.5 \times (1 - 0.428)) \xrightarrow{\text{CS algorithm}} \begin{pmatrix} 1.7 & 0.8 \\ 0.4 & 5.1 \end{pmatrix} \begin{matrix} 20.2 \\ 45.8 \end{matrix} \\
(5) \quad & \mu'_{HH}{}^2 = \mu'_{H0}\mu'_{0H} \exp(2 \times -2.3) \qquad \qquad \qquad 14.7 \quad 21.1 \\
(6) \quad & \mu'_{HL}{}^2 = \mu'_{H0}\mu'_{0L} \exp(2 \times -3.3) \\
(7) \quad & \mu'_{LH}{}^2 = \mu'_{L0}\mu'_{0H} \exp(2 \times -4.1) \\
(8) \quad & \mu'_{LL}{}^2 = \mu'_{L0}\mu'_{0L} \exp(2 \times -1.8)
\end{aligned}$$

Simulation 3: fix surplus, sex ratio and education gender gap

We keep the total number of men and the fraction of H -type men constant as observed in the census and we change: (i) the number of women to reflect the sex ratio (1.03) among cohorts active in the 1999 market and (ii) the fraction of H -type among women to reflect the gender gap in education (4.4p.p.) among cohorts active in the 1999 market.

$$\begin{aligned}
(1) \quad & \mu'_{HH} + \mu'_{HL} + \mu'_{H0} = 18.4 + 4.3(= 9.8 \times 0.442) \\
(2) \quad & \mu'_{LH} + \mu'_{LL} + \mu'_{L0} = 46.0 + 5.5(= 9.8 \times (1 - 0.442)) \\
(3) \quad & \mu'_{HH} + \mu'_{LH} + \mu'_{0H} = 12.8 + 3.8(= 9.5 \times 0.398) \\
(4) \quad & \mu'_{HL} + \mu'_{LL} + \mu'_{0L} = 21.5 + 5.7(= 9.5 \times (1 - 0.398)) \xrightarrow{\text{CS algorithm}} \begin{pmatrix} 1.7 & 0.8 \\ 0.4 & 5.1 \end{pmatrix} \begin{matrix} 20.2 \\ 45.8 \end{matrix} \\
(5) \quad & \mu'_{HH}{}^2 = \mu'_{H0}\mu'_{0H} \exp(2 \times -2.3) \qquad \qquad \qquad 14.5 \quad 21.3 \\
(6) \quad & \mu'_{HL}{}^2 = \mu'_{H0}\mu'_{0L} \exp(2 \times -3.3) \\
(7) \quad & \mu'_{LH}{}^2 = \mu'_{L0}\mu'_{0H} \exp(2 \times -4.1) \\
(8) \quad & \mu'_{LL}{}^2 = \mu'_{L0}\mu'_{0L} \exp(2 \times -1.8)
\end{aligned}$$

Appendix C Dynamic estimates of the marital surplus

We consider a simple version of the dynamic model in [Choo \(2015\)](#) with a two-dimensional type space (age and education categories), in which there is no risk of divorce. We further assume that individuals are myopic and believe that the market is at the steady state. Under these two assumptions, dynamic marriage gains to an $\{i, j\}$ marriage, $\Pi_{i,j}$, are given by

$$2\Pi_{i,j} = \ln\left(\frac{\mu_{i,j}}{m_i}\right) - \sum_{k=0}^{T_i} \ln\left(\frac{\mu_{i'(i,k),0}}{m_{i'(i,k)}}\right)^{\beta^k} + \ln\left(\frac{\mu_{i,j}}{f_j}\right) - \sum_{k=0}^{T_j} \ln\left(\frac{\mu_{0,j'(j,k)}}{f_{j'(j,k)}}\right)^{\beta^k} \quad (11)$$

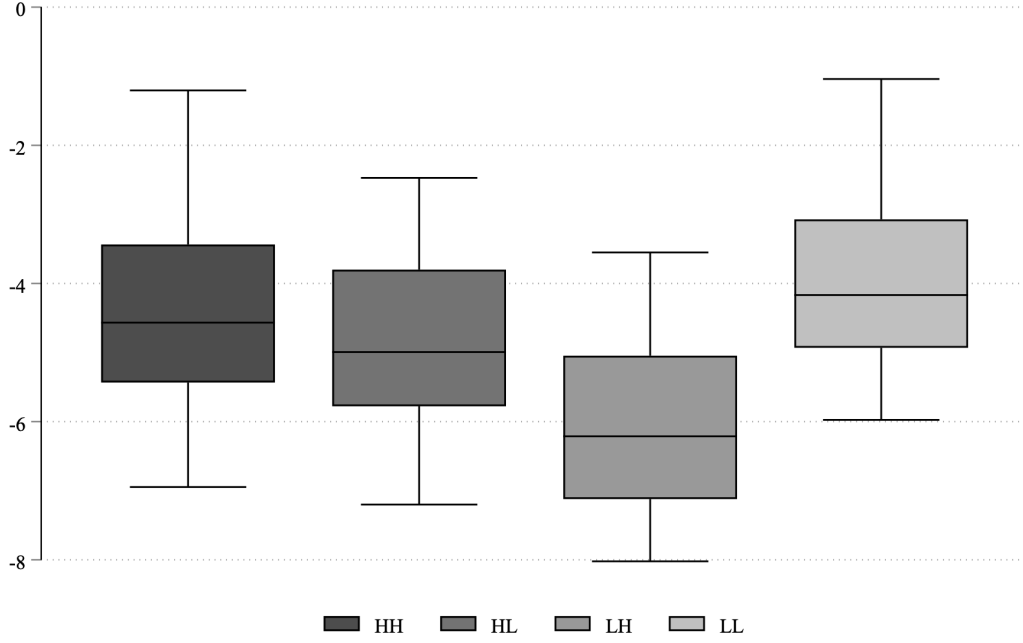
where $i = \{\text{age}, \text{educ}\}$, $j = \{\text{age}, \text{educ}\}$, $T_i = T - \text{age}_i + 1$ and $T_j = T - \text{age}_j + 1$. T is the last period for participating in the marriage market for both genders. $\mu_{i'(i,k),0}$ ($\mu_{0,j'(j,k)}$) is the number of men (women) of type i who remain single after period k . $m_{i'(i,k)}$ ($f_{j'(j,k)}$) is the total number of men of type i in the beginning of period k . β is the discount rate.

We aggregate age groups into 3-year intervals (19-21, 22-24, etc.) to ensure sufficient observations for each match type. We assume that individuals aged 43 and above no longer participate in the marriage market. We assume that a new marriage market is formed every three years (that is, the previous market clears and all remaining singles advance to the next age category and enter the new marriage market). Educational attainment is constant over time for a given individual and categorized as either high (H) or low (L), like in the static model. Following [Brandt et al. \(2018\)](#), we apply a tri-annual discount rate of $\beta = 0.94$.

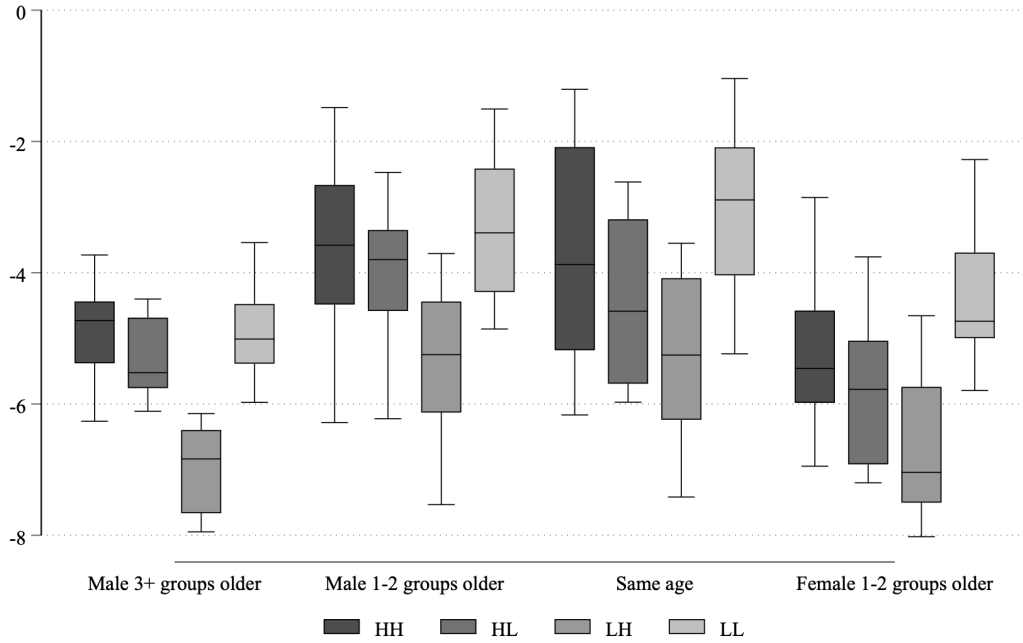
Figure [C.1a](#) plots the surplus distribution across all age combinations for the four match types defined by spousal education levels: HH , HL , LH , and LL . Unlike the static model, which has one value per match type, the dynamic model has several values per match type corresponding to different age groups. We use box plots to represent the min, max, first, second and third quartiles of the estimated surpluses for different age groups. We find the same ranking as in the static model: $LH < HL < HH < LL$. Figure [C.1b](#) plots the results disaggregated by spousal age difference. The most frequent age differences observed in the data are either same age group or male 1-2 groups older than female: they account for 87% of all matches. Again, the ranking across education types remains the same, irrespective of the age difference. Therefore, we do not seem to lose important information when we aggregate all age groups.

Figure C.1: Dynamic marital surplus by education types

(a) Dynamic marital surplus by education types



(b) Dynamic marital surplus by education types and age difference



Note: We pool age groups in 3-year intervals (19-21, 22-24, etc.) to have enough observations for each match type. Certain combinations occur with negligible frequency (e.g., low-educated men aged 19-21 with high-educated women aged 40-42). We restrict the sample to matches happening with a probability higher than 0.001 and use box plots to represent the min, max, first, second and third quartiles of the surplus. Figure (a) illustrates the surplus distribution across all age combinations for the four match types defined by spousal education levels. Figure (b) presents this distribution further disaggregated by spousal age differences.